

Single agent vs multi agent

Measured comparison / Local Ollama / qwen3:4b-instruct-2507-q4_K_M

ACTUAL EXECUTIONS

6/8

successful runs; 8/8 scheduled attempts

39.4 min

sum of measured matcher durations

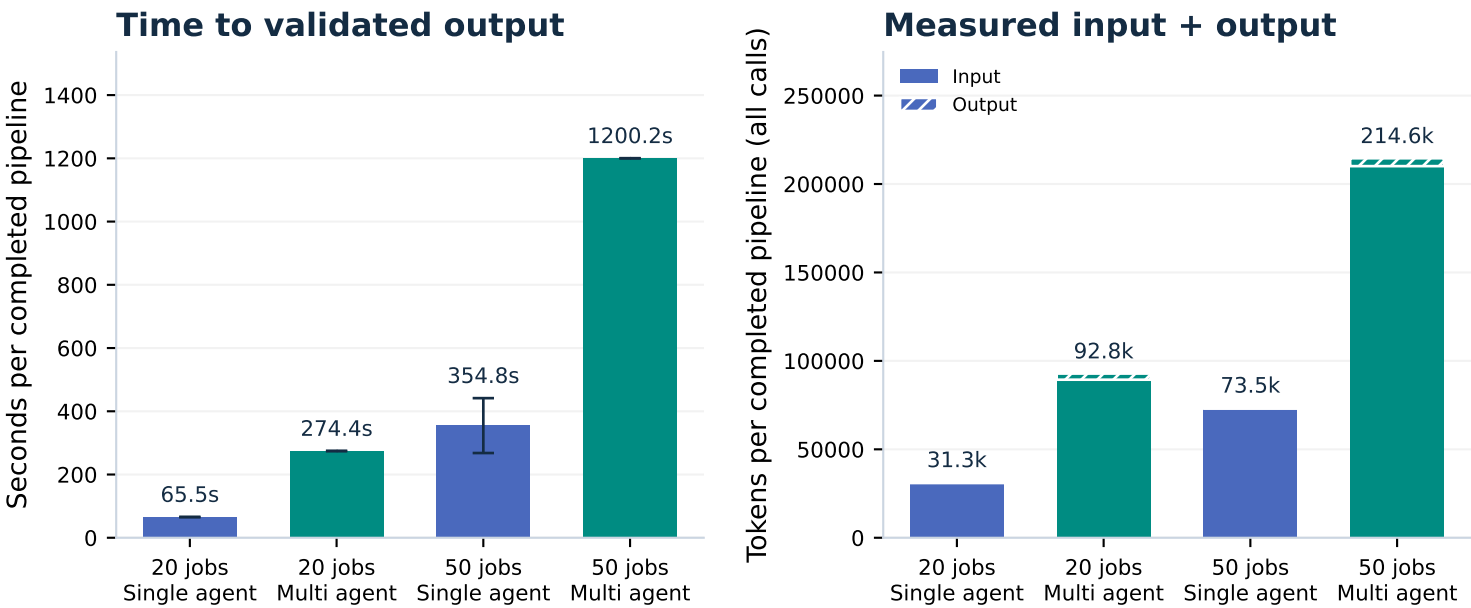
\$0.00

API token charges

Local inference has no API token billing. Electricity and hardware costs were not measured.

Jobs	Pipeline	OK / runs	Time / OK	Input / OK	Output / OK	USD / OK
20	Single agent	2 / 2	65.5s	30.9k	400	\$0.00
20	Multi agent	1 / 2	274.4s	89.4k	3.5k	\$0.00
50	Single agent	2 / 2	354.8s	73.1k	400	\$0.00
50	Multi agent	1 / 2	1,200.2s	210.1k	4.5k	\$0.00

Completed same-input pairs, multi / single: R01/20: 4.19x time; R01/50: 4.48x time



OK means a validated top-five output. Time shows the median of completed runs; whiskers show their minimum/maximum. Tokens show the mean per completed run, including all workers and retries. Failures are accounted for below and in CSV/JSON.

20 jobs, Single agent: 0 failed (0.0s / 0 tokens); 2 calls, 0 retries total.
20 jobs, Multi agent: 1 failed (9.7s / 2.4k tokens); 5 calls, 1 retries total.
50 jobs, Single agent: 0 failed (0.0s / 0 tokens); 2 calls, 0 retries total.
50 jobs, Multi agent: 1 failed (41.6s / 2.4k tokens); 5 calls, 1 retries total.

Recorded batch wall time: 39.44 min (includes loop/logging overhead).

Actual ranked outputs

R01 / 20 candidate jobs / repetition 1

SAME RESUME, PREFERENCES, JOBS AND JOB ORDER

Single agent

65.5s / 1 calls / \$0.00

1. J026

The resume demonstrates strong proficiency in Python, JavaScript, and full-stack development (React, Node.js, SQL), with direct experience...

2. J034

The candidate has hands-on experience with AI tools, including building an AI-powered note summarizer using Python and OpenAI API, and...

3. J006

The resume shows strong foundational skills in software development, including experience with C++, Java, Python, and full-stack...

4. J001

The candidate has a strong background in AI and machine learning through projects like an AI-powered note summarizer and experience with...

5. J009

The resume includes experience with full-stack development, AI integration, and prompt engineering, which aligns with the AI in...

Multi agent

274.4s / 3 calls / \$0.00

1. J026

The resume demonstrates proficiency in Python, SQL, React, Node.js, Docker, and Agile/Scrum, aligning with the required skills for the...

2. J032

The candidate has strong experience with Python, AI integration (OpenAI API), and prompt engineering, directly matching the required...

3. J031

The resume shows strong skills in Python, SQL, and web technologies, with experience building backend systems and handling data. The...

4. J030

The candidate has hands-on experience with Python, Flask, and OpenAI API, and has built web applications with React and Node.js. The Data...

5. J006

The resume shows strong programming skills in Python, Java, and C++, with experience in software development, system design, and AI...

Top-five overlap in this pair: 2/5 jobs. Explanations are shortened saved excerpts; full explanations are in runs.jsonl. Overlap measures agreement, not relevance.

WHAT THESE MEASUREMENTS COVER

Single agent makes one ranking request. Multi agent parses the resume, extracts all job requirements in one batch, and ranks using those structured outputs. Workers run sequentially; the controller adds no LLM call. Development benchmark: 2 resumes, 1 repetition(s) per condition, 8 attempts, 6 successes. Failed time/tokens count in batch totals and CSV/JSON. Throughput includes all attempt time. Failed runs: R02/20 Multi agent, R02/50 Multi agent. Saved errors: Unsupported resume evidence. Early exits are not completed-pipeline speeds. Human relevance labels are not used in this timing/cost report. Precision@5 and a winner on recommendation quality are not inferred from these sample rankings. Recorded platform: macOS-15.5-arm64-arm-64bit; Python 3.9.6. Ollama 0.35.0; configured context 131072 tokens (model limit 262144). Timings use one continuous local server session; model loading and cache state were not reset or standardized. The server had already been used for development before this benchmark. Price source (2026-10-06): Local Ollama inference: no API token billing; hardware/electricity costs unmeasured

Source: results/local_benchmark/runs.jsonl + manifest.json

8/8 scheduled attempts present. No extrapolated or hypothetical results.

Resume-to-job matching | CSCE 585 | Saved model responses and usage counters